

Goethe-Universität Frankfurt am Main
Faculty 12 Informatik und Mathematik

Bachelor Thesis Proposal

Approximating Subset-Performance Distributions
in Active Learning under Sample Constraints

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1 Introduction

Supervised Machine Learning relies heavily on labeled data, which is often expensive, time-consuming or scarce. Active Learning (AL) has been developed as a framework for reducing annotation costs by iteratively selecting the most informative samples for labeling [1]. Over the past decades, numerous query methods have been designed for various domains and have demonstrated strong performance in academic literature [2], [3].

1.1 Problem Statement

Although many query methods outperform random sampling in controlled settings, their performance in practical scenarios remains marginal and inconsistent [4]. Furthermore, evidence suggests the existence of specific subsets that could yield significantly higher performance than random subsets [5]. This gap between academic performance and practical effectiveness raises a fundamental question: why do query methods, despite years of research, struggle to reliably outperform random sampling?

One possible explanation lies in the structure of the subset performance landscape itself. If the distribution of model accuracy across all possible training subsets is highly non-uniform — with only a small fraction of subsets yielding high accuracy — then identifying those high-performance subsets becomes inherently difficult, regardless of the query strategy used.

In this thesis, AL serves as the motivation, whereas the primary research objective is the approximation of subset-performance distributions under limited samples. In this way, the prevalence of high-performance subsets can be quantified by approximating the distribution.

2 Research Questions and Hypotheses

2.1 Research Questions

RQ1: How is the performance of training subsets distributed across the entire combinatorial space of a controlled 2D toy dataset and larger benchmarks such as HoneyPot and Diverging Sine?

RQ2: Is the proportion of high-performing (i.g., accuracy ≥ 0.9) subsets sufficiently small (i.g., $\leq 5\%$).

RQ3: Can subset-performance distributions be reliably approximated from limited subset evaluations?

RQ4: How do different density estimation methods and sampling strategies affect the quality of this approximation?

2.2 Hypotheses

H1: The subset-performance distribution is significantly non-uniform, evidenced by a Wasserstein Distance significantly greater than zero between the empirical distribution and a uniform distribution over the same support.

H2: Only a small fraction (i.g., $\leq 5\%$) of subsets achieves high performance (i.g., accuracy ≥ 0.9)

H3: The performance distribution of training subsets can be approximated from a limited number of sampled subsets.

3 Research Design

The research is conducted in three stages. Each stage builds on the results of the previous one. First, the true subset-performance distribution is established on a toy dataset by exhaustive evaluation. Second, density estimation methods are applied to the toy dataset as approximation techniques under different sampling strategies and budgets. Third, the best-performing approximation strategy is tested on larger datasets where exhaustive enumeration is infeasible.

3.1 Stage 1: Exhaustive Evaluation

A ground-truth distribution is first established by evaluating all possible subsets from a controlled 2D toy dataset. Using a labeling budget of $k = 5$ out of $n = 30$ points, we conduct $C(30, 5) = 142,506$ independent training runs. The model consists of a single-layer neural network (no hidden neurons) trained with a fixed learning rate and optimizer to ensure reproducibility.

After exhaustive evaluation, several geometric and statistical metrics of the subsets are analyzed. This exploratory analysis aims to determine whether simple features can predict the accuracy of subsets. The Analysis is presented in the Appendix.

3.2 Stage 2: Approximation Strategies

Building on the insights from the exhaustive evaluation, we implement density estimation techniques to approximate the global distribution from a limited sample budget. Histogram estimation serves as the simplest baseline. Kernel Density Estimation (KDE) provides a smooth non-parametric method with well-studied convergence properties. Gaussian Mixture Model (GMM) provides a parametric method to compare with KDE. The sample efficiency of KDE and GMM is evaluated under various sampling rates (e.g., 1%, 5%, 10%, 25%). The quality of these approximations is quantified using Wasserstein Distance and KL Divergence against the ground-truth distribution.

3.3 Stage 3: Scaling to Larger Datasets

The best approximation strategy from stage 3 will be applied to larger datasets (e.g., Honeypot, Diverging Sine). Since exhaustive evaluation is computationally infeasible on larger datasets, approximation quality is assessed indirectly through convergence behavior — specifically, whether the estimated distribution stabilizes as sample size increases — and through bootstrap stability, which measures consistency across different random seeds. Wasserstein Distance is also applied here to quantify the approximation quality.

4 Thesis Outline

- Abstract
- Table of Contents
- **Chapter 1: Introduction**
 - 1.1 Motivation
 - 1.2 Problem Statement
 - 1.3 Structure of the Thesis
- **Chapter 2: Background**
 - 2.1 Active Learning
 - 2.2 Density Estimation under Limited Samples
 - * 2.2.1 Kernel Density Estimation (KDE)
 - * 2.2.2 Gaussian Mixture Model (GMM)
 - * 2.2.3 Stratified Sampling
 - * 2.2.4 Wasserstein Distance
 - * 2.2.5 Kullback-Leibler (KL) Divergence
 - 2.3 Problem Formulation and Mathematical Definitions
 - * 2.3.1 Dataset and Subset Space
 - * 2.3.2 Model performance on a subset
 - * 2.3.3 Performance distribution over subsets
 - * 2.3.4 High-performance subsets
 - * 2.3.5 Approximation objective
 - * 2.3.6 Quantiles
 - * 2.3.7 Approximation error
- **Chapter 3: Research Questions and Hypotheses**
 - 3.1 Research Questions
 - 3.2 Hypotheses
- **Chapter 4: Methodology**
 - 4.1 Exhaustive Evaluation
 - 4.2 Investigation of Subset Properties
 - 4.3 Approximation Strategies
 - 4.4 Scaling to Larger Datasets
- **Chapter 5: Results and Analysis**

- **Chapter 6: Discussion and Conclusion**

5 Timetable

Table 1: Timetable (9 Weeks)

Month	Week	Date	Date	Assignment
June	Week 01	01.06.2026	07.06.2026	Literature Review
	Week 02	08.06.2026	14.06.2026	Framework Setup
	Week 03	15.06.2026	21.06.2026	Experiment: Exhaustive Evaluation and Investigation of Metrics
	Week 04	22.06.2026	28.06.2026	Experiment: Design and Validation of Approximation Strategies
July	Week 05	29.06.2026	05.07.2026	Experiment: Scaling Approximation Strategies to Larger Datasets and Evaluation
	Week 06	06.07.2026	12.07.2026	Result and Analysis
	Week 07	13.07.2026	19.07.2026	Discussion and Conclusion
	Week 08	20.07.2026	26.07.2026	Proofreading, Formatting & Review
	Week 09	27.07.2026	02.08.2026	Submission Preparation & Final Hand-in

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